7 The total mass of a cyclist and his bicycle is 80 kg.

The cyclist starts from rest and rides down a hill without pedalling.

The vertical height of the hill is 50 m.

At the bottom of the hill, the speed of the cyclist is 20 m / s.

How much energy is wasted as the cyclist rides from the top of the hill to the bottom of the hill?

A less than 10 kJ

B between 10 kJ and 20 kJ

C between 20 kJ and 30 kJ

D over 30 kJ